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TASK	WIFI HACKING 101 REPORT

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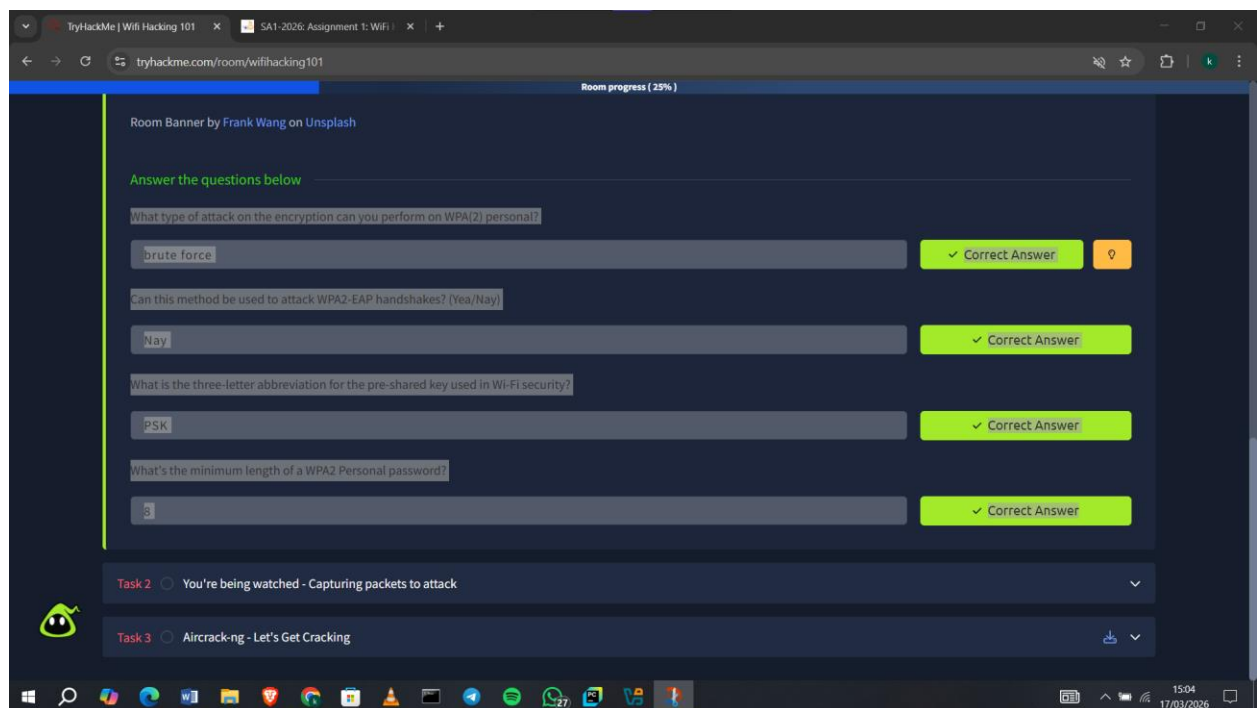
WIFI HACKING 101 REPORT

Introduction

In this module, I learned key WiFi concepts and authentication methods used in modern networks. I also gained hands-on experience using tools to capture packets and begin attacking WPA/WPA2 networks.

1.0 The Basics

I learned the key WiFi terms like SSID, BSSID, and the difference between WPA2-PSK and WPA2-EAP authentication. I also understood how the 4-way handshake works to verify devices without sharing the actual password. Additionally, I saw why WEP is insecure and how WPA2 security relies on both the password and network name, making attacks harder.

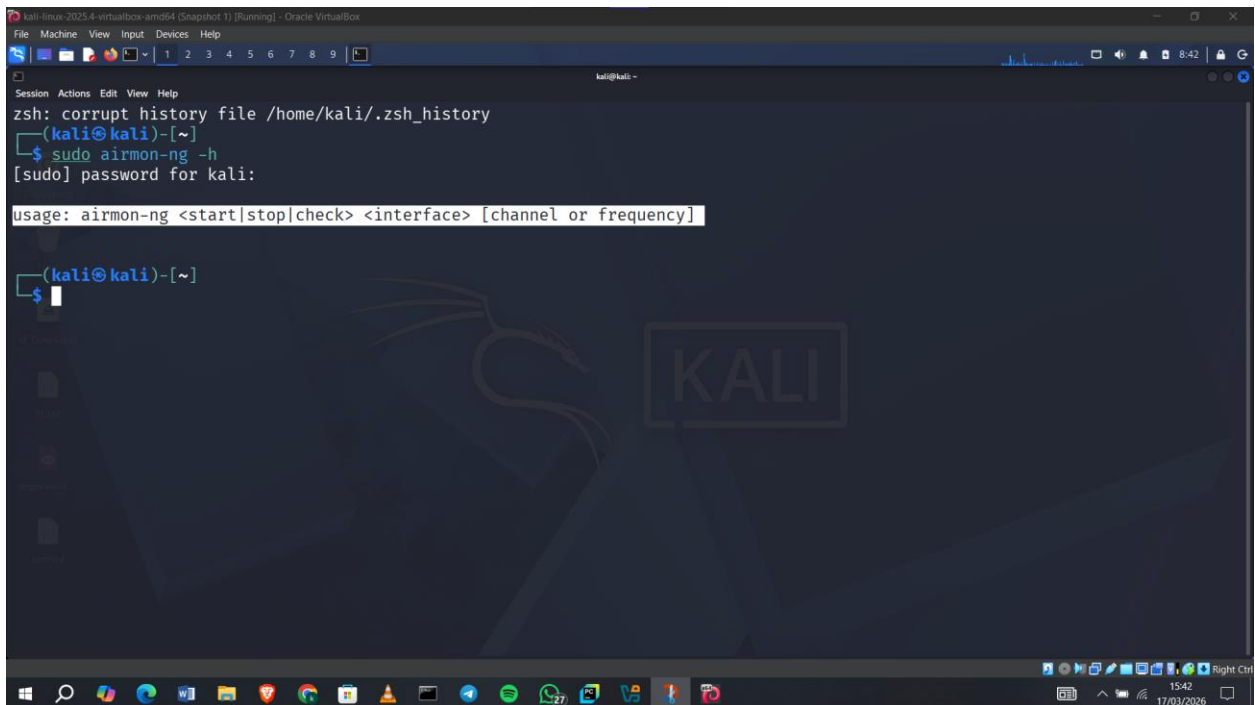


2.0 Capturing Packets to attack

I learned how to use the Aircrack-ng suite, especially tools like airmon-ng, airodump-ng, and aircrack-ng, to begin attacking WPA networks.

Question1: How do you put the interface “wlan0” into monitor mode with Aircrack tools?
(Full command)

Answer: **airmon-ng start wlan0**

A screenshot of a Kali Linux terminal window. The terminal shows the command 'sudo airmmon-ng -h' being executed. The output displays the usage for 'airmon-ng', which is: 'usage: airmmon-ng <start|stop|check> <interface> [channel or frequency]'. The terminal also shows a message from zsh about corrupting the history file and a password prompt for 'kali'. The background of the terminal has a faint Kali Linux logo.

```
kali@kali:~$ sudo airmmon-ng -h
[sudo] password for kali:
usage: airmmon-ng <start|stop|check> <interface> [channel or frequency]

(kali@kali)-[~]
$
```

Question2: What is the new interface name likely to be after you enable monitor mode?

Answer: **wla0mon**

Question3: What do you do if other processes are currently trying to use that network adapter?

Answer: **airmon-ng check kill**

```
kali-linux-2025.4-virtualbox-amd64 (Snapshot 1) [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
1 2 3 4 5 6 7 8 9 10
kali@kali ~
(kali@kali)~$ sudo airmmon-ng -h
[sudo] password for kali:

usage: airmmon-ng <start|stop|check> <interface> [channel or frequency]

(kali@kali)~$ airmmon-ng start wlan0
Run it as root

(kali@kali)~$ sudo airmmon-ng start wlan0

Found 1 processes that could cause trouble.
Kill them using 'airmon-ng check kill' before putting
the card in monitor mode, they will interfere by changing channels
and sometimes putting the interface back in managed mode

PID Name
631 NetworkManager

Requested device "wlan0" does not exist.
Run /usr/sbin/airmon-ng without any arguments to see available interfaces

(kali@kali)~$
```

Question4: What flag do you use to set the BSSID to monitor?

Answer: I ran **airodump-ng --help** and the flag was **--bssid**

```
kali-linux-2025.4-virtualbox-amd64 (Snapshot 1) [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
1 2 3 4 5 6 7 8 9 10
kali@kali ~

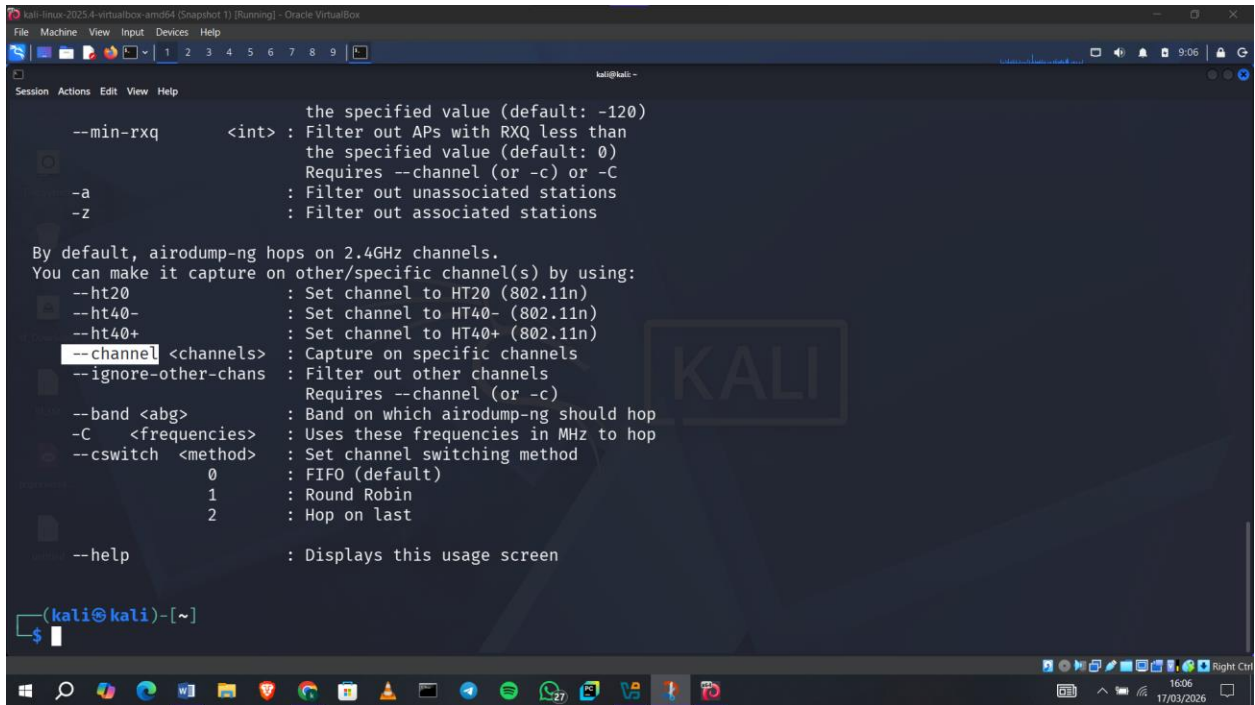
fixed channel <interface>: -1
--write-interval <seconds> : Output file(s) write interval in seconds
--background <enable> : Override background detection.

Filter options:
--encrypt <suite> : Filter APs by cipher suite,
                  you can pass multiple --encrypt options
--netmask <netmask> : Filter APs by mask
--bssid <bssid> : Filter APs by BSSID,
                you can pass multiple --bssid options
--essid <essid> : Filter APs by ESSID,
                you can pass multiple --essid options
--essid-regex <regex> : Filter APs by ESSID using a regular
                      expression
--min-packets <int> : Minimum AP packets recvd before
                   displaying it (default: 2)
--min-power <int> : Filter out APs with PWR less than
                  the specified value (default: -120)
--min-rxq <int> : Filter out APs with RXQ less than
               the specified value (default: 0)
               Requires --channel (or -c) or -C
-a : Filter out unassociated stations
-z : Filter out associated stations

By default, airodump-ng hops on 2.4GHz channels.
You can make it capture on other/specific channel(s) by using:
```

Question5: And to set the channel?

Answer: **--channel**



```
kali@kali:~$ aircrack-ng --help
--min-rxq <int> : the specified value (default: -120)
                  : Filter out APs with RXQ less than
                  : the specified value (default: 0)
                  : Requires --channel (or -c) or -C
-a : Filter out unassociated stations
-z : Filter out associated stations

By default, airodump-ng hops on 2.4GHz channels.
You can make it capture on other/specific channel(s) by using:
--ht20 : Set channel to HT20 (802.11n)
--ht40- : Set channel to HT40- (802.11n)
--ht40+ : Set channel to HT40+ (802.11n)
--channel <channels> : Capture on specific channels
--ignore-other-chans : Filter out other channels
                  : Requires --channel (or -c)
--band <abg> : Band on which airodump-ng should hop
-C <frequencies> : Uses these frequencies in MHz to hop
--cswitch <method> : Set channel switching method
                  : 0 : FIFO (default)
                  : 1 : Round Robin
                  : 2 : Hop on last
--help : Displays this usage screen

(kali@kali)-[~]
$
```

Question6: And how do you tell it to capture packets to a file?

Answer: **-w**

3.0 Aircrack-ng

Question1: What flag do we use to specify a BSSID to attack?

Answer: **-b**

Question2: What flag do we use to specify a wordlist?

Answer: **-w**

Question3: How do we create a HCCAPX in order to use hashcat to crack the password?

Answer: **-j**

Question4: Using the rockyou wordlist, crack the password in the attached capture. What's the password?

Answer: I ran **sudo aircrack-ng -b 02:1A:11:FF:D9:BD -w /usr/share/wordlists/rockyou.txt NinjaJc01-01.cap** and found the password as **greeneggsandham**

```
kali@kali: /media/sf_Downloads/Captures_1578171018678
Aircrack-ng 1.7
[00:01:18] 124414/14344392 keys tested (1605.44 k/s)
Time left: 2 hours, 27 minutes, 37 seconds 0.87%
KEY FOUND! [ greeneggsandham ]

Master Key      : 71 5F 17 D1 D7 9E 70 4D 6E 2E 9C AD 46 F5 45 F5
                  AF 5E 43 48 16 F9 5B AA 14 8F 39 AA FC 5E EB 3B

Transient Key   : 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
                  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
                  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
                  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00

EAPOL HMAC     : 9A 6A 56 EE E4 4E 42 A3 14 71 26 9F E0 E2 93 04

(kali@kali)-[/media/sf_Downloads/Captures_1578171018678]
```

Question5: Where is password cracking likely to be fastest, CPU or GPU?
Answer: **GPU**

TryHackMe | Wifi Hacking 101

SA1-2026: Assignment 1: Wifi

airmon-ng [Aircrack-ng]

tryhackme.com/room/wifihacking101/congratulations?step=records

Share your win with your peers

Your progress can inspire others in your community.



Wifi Hacking 101 completed!

kenmuirun07 completed another room on their cyber security journey.

Completed tasks

3

Points earned

128

Streak

1

Share on social media

Share your win before starting the next room

LinkedIn

WhatsApp

Telegram

Twitter / X

Facebook

Reddit

Go to dashboard

Conclusion

Overall, this module helped me build a practical understanding of WiFi attacks, especially capturing and cracking WPA/WPA2 handshakes.